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Three Transiting Candidates around Faint M Dwarfs from an Agent-Driven Search of *TESS* Full-Frame Images

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ABSTRACT

We report three transiting planet candidates around M dwarfs with *TESS* magnitudes 13.8–14.8, identified in an independent box-least-squares search of SPOC full-frame image (FFI) cutouts of 38,620 Sector 1 stars with $12 < T < 15$ and $R_\star < 0.6\,R_\odot$ that were neither 2-minute targets nor *TESS* Objects of Interest. Each signal is recovered independently in every available sector (3–9 sectors, 2018–2026), is located on the target in noise-normalized difference images, shows consistent odd and even depths and no secondary eclipse, and is fit with a physical transit model. TIC 389723320 hosts a $4.6\,R_\oplus$ candidate at $P = 3.927167$ d; TIC 114949540 a $4.35\,R_\oplus$ candidate at $P = 2.739176$ d; and TIC 259719966 a $0.94\,R_{\rm Jup}$ companion at $P = 11.077205$ d whose nature requires a mass measurement. None appears in the TOI, Community Candidate, or confirmed-planet lists. The pipeline was built and operated with an AI coding agent under human direction; code and per-candidate vetting reports are public.

Keywords: Exoplanet detection methods (489) — Transit photometry (1709) — M dwarf stars (982) — Astronomy software (1855)

MOTIVATION

The *TESS* project's own FFI transit searches have concentrated on stars brighter than $T \approx 13.5$ (Huang et al. 2020; Kunimoto et al. 2022), leaving the fainter M-dwarf population, where a given planet produces the deepest transits, comparatively under-searched. We targeted that population directly: M dwarfs ($T_{\rm eff} < 4000$ K, $R_\star < 0.6\,R_\odot$) with $12 < T < 15$, excluding 2-minute targets, TOIs, Community Candidates, and known planet hosts. For Sector 1 this yields 40,328 unique stars, of which the 38,620 with contamination ratio below 1 were searched.

DATA AND SEARCH

For each star we extracted a 15×15 pixel cutout from the SPOC calibrated FFI sector cubes hosted on AWS (astrocut; Brasseur et al. 2019), defined a threshold aperture, summed the aperture flux, and removed shared systematics by regressing against the leading principal components of the non-aperture pixels (lightkurve; Lightkurve Collaboration et al. 2018). After normalization, a 33-hour Savitzky–Golay flatten, and a point-to-point-jump filter for pointing excursions, we ran a box least squares search (Kovács et al. 2002) over 0.3–13 d with durations 0.5–6 h. Candidates required S/N ≥ 9, depth ≥ 3000 ppm, at least two transits, duration no more than 1.5× the maximum allowed by the TIC stellar density (Stassun et al. 2019), duty cycle below 15%, odd and even depths consistent within 3σ, and no secondary eclipse at phase 0.5 above 3σ. Twenty-two stars passed.

VETTING

Each flagged star was tested against Gaia DR3 (Gaia Collaboration et al. 2023) neighbors within 2 pixels, with the eclipse depth each neighbor would need to reproduce the signal; a flux-weighted centroid shift between in- and out-of-transit cadences; and a difference image (out-of-transit mean minus in-transit mean, per-frame plane-subtracted, divided by its per-pixel noise) whose peak locates the source of the dip (Bryson et al. 2013; Twicken et al. 2018). Of the 22, seven were located on a neighbor, one was a spotted rotator, one an unresolvable 1″ blend, and three showed

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anomalous centroid behavior. Every survivor was then searched independently in each sector in which *TESS* observed it (sectors from tess-point; Burke et al. 2020) and the sectors stitched to refine the period. Two high-S/N single-sector detections failed to repeat and were dropped. Three candidates passed all tests.

CANDIDATES

Transit parameters were obtained with `batman` (Kreidberg 2015) and `emcee` (Foreman-Mackey et al. 2013), fitting epoch, period, $R_p/R_\star$, $a/R_\star$, impact parameter, and quadratic limb darkening with Gaussian priors (0.3 ± 0.1), supersampling each sector to its exposure time. Stellar parameters are from TIC v8. Figure 1 shows the phase-folded data and models.

TIC 389723320 ($T = 14.52$, $T_{\rm eff} = 3271$ K, $R_\star = 0.37\,R_\odot$, $d = 126$ pc). Six sectors (1, 28, 68, 95, 105, 106), 36 transits. $P = 3.927167 \pm 0.000001$ d, $T_0 = 2460899.3895 \pm 0.0005$ BJD, depth $12{,}850 \pm 690$ ppm, duration 1.50 ± 0.06 h, $b = 0.74$, $R_p = 4.62 \pm 0.13\,R_\oplus$, $T_{\rm eq} \approx 565$ K. Odd minus even depth 1100 ± 520 ppm (2.1σ). Difference images in sectors 68, 95, and 105 peak on the target at 14, 14, and 18σ. Transit-derived stellar density $4.1^{+1.5}_{-1.0}\,\rho_\odot$ versus 7.0 from the TIC, consistent within the b–$a/R_\star$ degeneracy.

TIC 114949540 ($T = 14.78$, $T_{\rm eff} = 3320$ K, $R_\star = 0.42\,R_\odot$, $d = 168$ pc). Three sectors (1, 27, 105), 25 transits. $P = 2.739176 \pm 0.000001$ d, $T_0 = 2461212.9505 \pm 0.0007$ BJD, depth 8930 ± 710 ppm, duration 1.55 ± 0.07 h, $b = 0.53$, $R_p = 4.35 \pm 0.17\,R_\oplus$, $T_{\rm eq} \approx 670$ K. Odd minus even -240 ± 450 ppm. Difference images in sectors 27 and 105 peak on the target at 15 and 8σ.

TIC 259719966 ($T = 13.80$, $T_{\rm eff} = 3443$ K, $R_\star = 0.49\,R_\odot$, $d = 147$ pc). Nine sectors (1, 27, 67, 94, 95, 101–104), 20 transits. $P = 11.077205 \pm 0.000001$ d, $T_0 = 2460877.8231 \pm 0.0002$ BJD, depth $38{,}300 \pm 1200$ ppm (undiluted sectors), duration 2.37 ± 0.03 h, $b = 0.55$, $R_p = 10.58 \pm 0.17\,R_\oplus$ ($0.94\,R_{\rm Jup}$), $T_{\rm eq} \approx 400$ K. Odd minus even -230 ± 370 ppm; secondary 20 ± 540 ppm. Difference images in sectors 94 and 104 peak on the target at 33 and 37σ. A Gaia neighbor 2.1 px away (27% of the target's G-band flux) enters the automatic aperture in three sectors and dilutes the depth there; the sense of that effect (shallower when included) independently places the dip on the target. With a radius near Jupiter's, the companion may be a giant planet, a brown dwarf, or a star near the hydrogen-burning limit; its radial-velocity semi-amplitude would exceed 100 m s^{-1} for any stellar-mass companion.

A residual search of the stitched light curves of the first two systems, with the known transits masked, found no additional signal above ∼1000 ppm with three or more transits at $P < 30$ d.

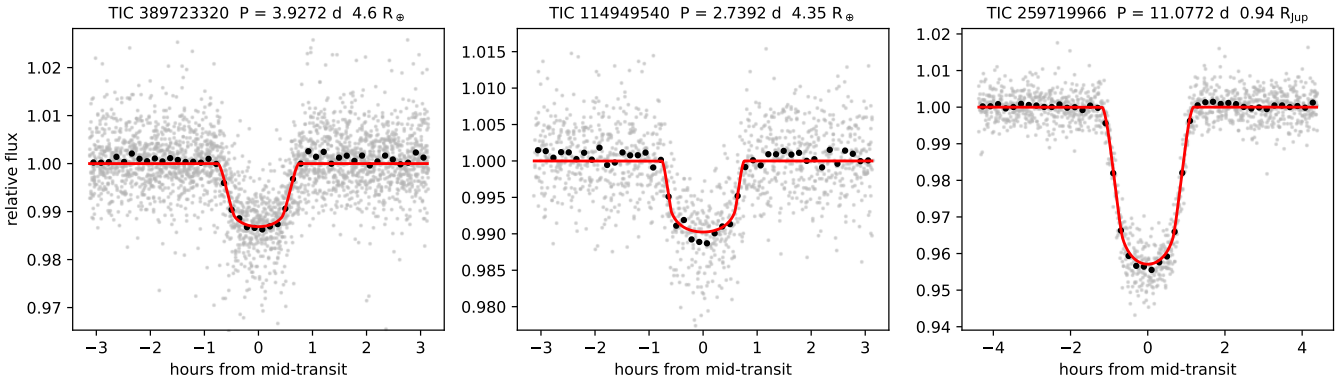


Figure 1. Phase-folded *TESS* FFI photometry (all sectors, gray) with the posterior-median transit model (red) for TIC 389723320 (left), TIC 114949540 (center), and TIC 259719966 (right). Binned points in black.

CAVEATS AND AVAILABILITY

All three hosts are faint for radial-velocity follow-up; TIC 259719966 is the most tractable. None of the candidates is a TOI, Community Candidate (ExoFOP list of 2026 October 24), or known planet host; none has an ADS literature match; and none lies within 30″ of an entry in the DIAmante FFI candidate catalog (Montalto et al. 2020), the QLP faint-star TOI release (Kunimoto et al. 2022), the *TESS* eclipsing binary catalog (Prša et al. 2022), or the Gaia DR3 variability classification. The pipeline (target selection, parallel FFI sweep, vetting, multi-sector recovery, MCMC fitting, and report generation) was written and operated with Claude, an AI coding agent, directed by the author,

who has no prior background in exoplanet science; we note this because the result bears on how such searches can be scaled. Code, pitfalls log, and per-candidate vetting reports are available at REPOSITORY-URL.

This research made use of data from the *TESS* mission, obtained from MAST at STScI, and of the NASA Exoplanet Archive and ExoFOP, operated by Caltech/IPAC under contract with NASA. It made use of data from the ESA mission Gaia, processed by the Gaia DPAC.

Software: astrocut (Brasseur et al. 2019), astropy, astroquery, batman (Kreidberg 2015), emcee (Foreman-Mackey et al. 2013), lightkurve (Lightkurve Collaboration et al. 2018), tess-point (Burke et al. 2020), Modal

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